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Received 4 February 2020
Revised 14 April 2020
25 April 2020
Accepted 4 May 2020

Development of a decision support system for assessing the supply chain risk mitigation strategies: an application in Indian petroleum supply chain

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Abstract

Purpose – The purpose of this paper is to develop a decision support system (DSS) to assist supply chain (SC) risk managers to select a suitable *risk management* (RM) strategy and expedite the implementation of corresponding RM enablers. The relationship between RM strategies and RM enablers is explored by identifying the underlying factors between them, which is further used to build the DSS.

Design/methodology/approach – The DSS is built by integrating heterogeneous techniques. A *systematic review* approach is employed to explore both *proactive* and *reactive* RM enablers, and they are further mapped to various RM strategies by using *correspondence analysis* (CA). An *in-depth interview* is conducted to develop the rules for constructing the decision system. A rule-based *fuzzy inference system* (FIS) is utilized to counteract the uncertainty involved in the decision variables. The efficacy of the proposed DSS is demonstrated by considering two conjectural scenarios in the case of Indian petroleum SC (IPSC).

Findings – The results reveal three primary underlying factors between the risk mitigation strategies viz. SC managers' *preparedness* to face risk, organization's *resource capability* to deal with risk and the *sophistication of the implementation* of the RM enablers; with explained variances of 37%, 29% and 22%, respectively. *Risk avoidance* strategy comprises of RM enablers such as *supplier evaluation, technology adaption, information security,* etc. Whereas, the *risk-sharing* strategy includes *revenue sharing, insurance, collaboration, public-private-partnership, etc.* as essential RM enablers. The DSS recommends *risk-mitigation* and *risk-sharing* as effective RM strategies for the IPSC under the considered scenarios.

Research limitations/implications – This paper develops a decision support framework for recommending an effective risk mitigation strategy and outranking the corresponding enablers. The study explicitly focuses on the risk mitigation step of the supply chain risk management (SCRM) process. Pre- and post-risk mitigation steps of the SCRM process, such as risk assessment and risk monitoring are beyond the scope of this research.

Originality/value – The operational procedure of the proposed DSS is explained by considering a real-life case of petroleum SC in the Indian scenario. The unique contributions of this study are presented as *theoretical implications* and *managerial propositions* in the context of a developing country.

Keywords Supply chain risk management, Risk mitigation, Decision support system

Paper type Research paper

Journal of Manufacturing Technology Management
Vol. 32 No. 2, 2021
pp. 506-535
© Emerald Publishing Limited
1741-038X
DOI 10.1108/JMTM-02-2020-0035

The authors acknowledge the two anonymous reviewers for their constructive feedback, which helped to enhance the quality of this paper.

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1. Introduction and background:

The ever-growing global competition and enormously uncertain business environment are causing a modern supply chain (SC) to become *vulnerable* and hence more susceptible to risk. The continuous attempt to achieve profitability and the emergence of recent manufacturing practices (like *globalization, single sourcing, lean manufacturing, etc.*) are pushing modern SCs toward risk (Tarei *et al.*, 2018). The risk in SC leads to serious repercussions like inferior product quality, devastation of property and machinery, loss of firm's reputation, delivery delays (Cousins *et al.*, 2004), manufacturing disruption (Zhang *et al.*, 2018), conflict among various shareholders (Craighead *et al.*, 2007) and sharp decline in firm's share price (Hendricks and Singhal, 2005). A practical implementation of supply chain risk management (SCRM) improves not only the financial performance of the firm (Fan *et al.*, 2017) but also the manufacturing effectiveness and overall performance of SC (Thun and Hoenig, 2011). The process of SCRM starts with the identification of risk drivers, assessment of the critical drivers and implementation of mitigation practices. In this regard, most of the research has been conducted toward identification and assessment of the potential risk drivers as compared to the risk mitigation phase (Ghadge *et al.*, 2013). Hence, we explicitly focus on the "Risk mitigation" stage of the SCRM cycle and discuss the research limitations about the risk mitigation phase in the following section.

Risk mitigation is one of the crucial phases of the SCRM cycle, which involves the selection of an effective risk management (RM) strategy and implementation corresponding RM enablers to nullify the adverse impact of risk. In most of the existing literature, RM *enablers* (also referred to as *risk responses*, *risk protection strategies*, *mitigation tactics*, *redesign principles, etc*) and RM *strategies* are used interchangeably (Vlajic *et al.*, 2012). We, however, argue that RM enablers are actual SC practices that are implemented at the operational level in the real-world business environment to reduce the overall riskiness of the SC (definitional significance of *RM strategy* is highlighted in next paragraph). We prefer to choose the term as *RM enablers* (abbreviated as RMEs) and use it consistently hereafter. These RMEs are primarily categorized into two groups, viz. proactive (preventive) RMEs and reactive (recovery) RMEs. Proactive RMEs show their impact beforehand and are primarily responsible for reducing the probability of occurrence of a risk event (Sheffi and Rice, 2005; Meng, 2019). On the contrary, reactive RMEs are rather effect-oriented, which can only show an impact after the occurrence of the risk event and are responsible for suppressing the adverse effects of an unforeseen event (Thun and Hoenig, 2011). The consideration of both proactive and reactive RMEs in a single model in the domain of SCRM is relatively scanty (Ho *et al.*, 2015; Rajagopal *et al.*, 2017).

Roh *et al.* (2014) defined SC strategy as a plan, which is decided by the top executives (following the corporate strategy) of a firm which precedes SC practices. The definition of RM strategies (abbreviated as RMSs) used in this research is adapted from the above source. Hence, we conceive and define RMS as a plan (not action), which is decided by the top executives (aligned with the corporate strategy) of a firm to counteract the future (adverse) impact of the risk. Though numerous attempts have been devoted to evaluate various RM strategies under different operational scenario (Aqlan and Lam, 2015; Qazi *et al.*, 2018), the existing literature has failed not only to highlight the distinction between RMEs and RMSs but also to provide an established relationship between RMEs and RMSs for the treatment of risk.

In the era of industry 4.0, the integration of modern (intelligent) tools and real-time information processing systems help SC to achieve organizational competency. This brings a new opportunity for managers to react to and manage SC risk in a shorter time frame (Güller *et al.*, 2015). The embedded intelligence helps SCM to move from supporting decisions to assigning them and, ultimately, to predict which decisions need to be made (Butner, 2010). Some recent studies (Schlüter and Sprenger, 2016; Kirazli and

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Moetz, 2015) on digitalization in SCRM have focused on identification of potential risks through implementation of technologies. Saridakis *et al.* (2018) highlighted the assessment of RMSs and the implementation of corresponding RMEs to nullify the devastation effect of risk. In the context of SCRM, limited studies have been performed, which primarily focus on the development of a decision support system (DSS) to assist SC risk managers.

Responding to all the research limitations mentioned above, we frame the following research questions (RQ) for the current study.

RQ1. What is the importance of various proactive and reactive RMEs, and how can they be considered simultaneously in an SCRM model?

RQ2. What is the significance of different RMSs, and how can they be linked with the corresponding RMEs?

RQ3. How can a DSS be developed which can assist the SC managers in making decisions about the selection of an RMS and hence the implementation of RMEs?

The remainder of this paper is structured as follows: Literature related to the risk mitigation phase of the SCRM process and the techniques adapted to build the DSS are reviewed in Section 2. Section 3 deals with the conceptualization and development of the DSS. A demonstration of the proposed DSS in the context of a case industry is provided in Section 4. The unique contributions of the study are discussed in Section 5. Finally, the research paper ends with conclusions and future research directions in Section 6.

2. Literature review and methodologies adopted:

In this section, we provide a brief overview of relevant literature and (review of) various methodologies that are adopted from multiple disciplines to build the DSS.

2.1 Supply chain risk management process

Baryannis *et al.* (2018b) generalized the definition of SCRM (after surveying prior definitions): *"as a process that encompasses the collaborative and coordinated efforts of all parties involved in a SC to reduce vulnerability and increase robustness and resilience of the supply chain, ensuring profitability and continuity."* The various stages of SCRM process consist of identification, assessment, mitigation and monitoring of risks (Baryannis *et al.*, 2018b). Risk classification and identification have been exhaustively discussed in the SCRM literature (Christopher and Peck, 2004; Wagner and Bode, 2008; Ghadge *et al.*, 2013; Tarei *et al.*, 2018). There is relatively less research that examines the effectiveness of alternative SCRM strategies in the different contextual environment (Chang *et al.*, 2015; Ayağ *et al.*, 2013). Hence, the current research primarily focuses on the *risk mitigation* step of the SCRM process and summarizes the relevant articles in Tables 1.

There are a few limitations to the studies above. First, very few modeling efforts have been attempted considering both proactive and reactive RMEs in a single framework. Second, distinct studies have been done to assess the RMEs and RMSs, but to the best of our knowledge no research has been conducted to link the relationship between RMEs and corresponding RMSs. As shown in Tables 1, the past decade has experienced accelerated growth of literature by heterogeneous methodologies (conceptual, qualitative, quantitative and interdisciplinary) in the domain of SCRM. Each method has its pros and cons, for instance, qualitative methods often provide a false sense of certainty and are criticized for their simplicity. Quantitative methods, on the other hand, are constrained by information quality, time and resources (Wang *et al.*, 2017). Hence, there is a need to build an integrated DSS, comprising the benefits of both qualitative and quantitative tools.

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We address the aforementioned research issues by proposing a DSS, which shall accelerate the decision-making process in the domain of SCRM. The next section deals with the review of proposed DSSs in the existing research along with various tools and techniques, which are used for building the DSS for the present research.

2.2 Tools used for building the DSS

The usage of DSS in the domain of SCRM is relatively scanty. Kumar and Viswanadham (2007) used a DSS adopting the case-based reasoning approach to evaluate the preventive and interceptive actions against construction SC risk.

In the following section, we review the usage of various tools and techniques (in the context of SC), which are used for the construction of the proposed DSS.

2.2.1 Systematic review:. We adopt a *three-level-keyword* based *systematic review* approach to explore various (*both proactive and reactive*) RMEs and RMSs from a wide variety of industries. The first level corresponds to keywords like *supply chain risk* OR *supply chain resilience* OR *supply chain vulnerability* to capture the context of "SC Risk". To capture the context of "Risk Mitigation", the second level keywords include *mitigation* OR *management* OR *response* OR *treatment* OR *protection*. The final level corresponds to keywords like *enabler* OR *strategy* OR *strategies* OR *antecedents* OR *practice* etc. All the levels are conjoined by "AND" set with each other and applied to the [Title, Abstract and Keyword] field only. We use leading bibliographic databases like *Web of Science, Scopus, Emerald, Taylor and Francis, Springer, etc.* The search results are further improved by imposing additional filters like journal articles written in *English* and excluding *conferences, book series, commercial articles,* etc.

The search results obtained from the review are used to explore various RMEs across heterogenous case industries (summarized in Table 2). The table excludes some review papers and articles which are based on the SC risk identification and assessment phase (like Baryannis *et al.*, 2018b; Tarei *et al.*, 2018; Ho *et al.*, 2015) and merely focuses on different proactive and reactive RMEs (as per the definition in Section 1). Since SCRM as a domain gained momentum after the year 2000 (Baryannis *et al.*, 2018b), it can be observed that all the articles appear in the range of the year 2000–2018.

RMSs (as per our proposed definition in Section 1) are alienated from those of RMEs and classified as shown in Table 3. We have considered and summarized the definitional significance of six RMSs, which appeared frequently in the prior literature.

2.2.2 Correspondence analysis. Correspondence analysis (CA) is a multivariate statistical technique, used to establish the relationship between two sets of nominal variables (RMEs and RMSs in this case) in the form of a *perceptual* map. The position of the variables on the perceptual map is defined by the weighted chi-square distance between them. Loureiro *et al.* (2015) applied CA to explore the relationship between the literature stream of *strategies* and *supply chain management*. Moreno *et al.* (2016) used CA to establish the correlation between the causes of *hazards* and *consequences* in biogas production. Oliva *et al.* (2016) examined the relationship between agents and event for enterprise RM by CA. Interested readers are suggested to refer to Greenacre (2017) for the detailed mathematical steps of CA.

2.2.3 Fuzzy inference system. Decision-making related to SCRM always involves the accumulation of tacit knowledge and expertise of key stakeholders. To automate the decision-making cycle, tacit knowledge has to be encoded and transformed into a machine-interpretable form. A popular method is a rule-based representation (with case-based being a common substitute). The rules are not only easy to construct by simple *if-then* formalization but also easily interpretable by both humans and machines (Baryannis *et al.*, 2018a). The FIS involves a process of choosing one or more (alternatives) solutions from a finite set of

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Table 1. Summary of approaches for SC risk mitigation

Authors	Qualitative	Quantitative		Empirical	Decision-making	Interdisciplinary	Case industry
		Analytical	Simulation				
Faisal *et al.* (2007)					ANP	SCOR	Information system
Oke and Gopalakrishnan (2009)	Interview			Multiple case			Retail
Sidola *et al.* (2011)			SD				–
Carvalho *et al.* (2012)			DES	Case study			Automotive
Wieland *et al.* (2012)				SEM			Manufacturing
Elleuch *et al.* (2014)	FMEA		DES		AHP		Pharmaceutical
Micheli *et al.* (2014)		Stochastic ILP				Fuzzy	Baby care products
Zhang and Fan, 2014		LP					–
Chang *et al.* (2015)	Conceptual						–
Aqlan and Lam (2015)	Bow Tie	Stochastic ILP					Server manufacturing
Lam *et al.* (2015)					AHP	CBR	Warehouse
Rajesh and Ravi (2015)					DEMATEL	Grey theory	Electronics
Ge *et al.* (2016)		Optimization	Numerical				Food (Wheat)
Giannakis and Papadopoulos (2016)	FMEA						Textile
Ramkumar *et al.* (2016)	SWOT				ANP	Fuzzy	E-procurement
Prakash *et al.* (2017)	FMEA				ISM		Dairy
Wang *et al.* (2017)	Delphi				TOPSIS	Fuzzy	Kitchen appliance

Note(s): AHP, analytical hierarchical process; ANP, analytical network process; CBR, case based reasoning; DEMATEL, decision-making trial and evaluation laboratory; DES, discrete event simulation; FMEA, failure mode and effect analysis; ILP, integer linear programming; ISM, interpretive structural modeling; SD, system dynamics; SEM, structural equation modeling; TOPSIS, technique of order preference by similarity to ideal solution

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Risk mitigation Practices/enablers ⬇	Authors (Year)																																		
	Ali *et al.* (2017)	Carbonara & Pellegrino (2017)	Chopra & Sodhi (2004)	Blackhurst *et al.* (2005)	Christopher & Peck (2004)	Christopher *et al.* (2011)	Christopher & Holweg (2011)	Dallaheway *et al.* (2017)	Faisal *et al.* (2006)	Fernendas *et al.* (2010)	Ghadge *et al.* (2012)	Irene Kilubi (2016)	Lawstre *et al.* (2012)	Machado *et al.* (2018)	Olson & Swenseth (2014)	Rajesh & Ravi (2015)	Rajesh *et al.* (2015) IJPR	Rajippal *et al.* (2017)	Sedhi & Lee (2007)	Soni *et al.* (2014)	Stecke & Kumar (2009)	Tang (2006)	Tang & Tomlin (2008)	Takamuthabwa *et al.* (2017)	Urciuoli *et al.* (2014)	Vlajic *et al.* (2012)	Wang, Tiwari & Chen (2017)	Zsidisin, Panelli & Upton (2000)
Revenue Management*															✓		✓			✓		✓				✓		
Dynamic assortment planning																✓	✓				✓	✓						
Silent production rollover																	✓					✓				✓		
Aggregate (Pool) demand			✓												✓		✓									✓		
Multiple customer account																		✓										
Monitor customer trend																					✓							
(Accurate) demand forecast																✓												
Flexible labor							✓																					
Revenue sharing									✓													✓						
Flexible transportation										✓							✓				✓	✓		✓				
Technology adaptation							✓									✓										✓		
Weather forecasting																					✓							
Adaptive capability	✓																			✓						✓		
Responsiveness (Pricing)			✓										✓			✓	✓						✓				✓	
Supply Chain visibility	✓			✓	✓	✓					✓			✓		✓			✓	✓	✓			✓				
Supply Chain velocity	✓				✓	✓										✓												
Strategic risk planning									✓								✓											
(Information) security	✓								✓								✓					✓			✓		✓	
Integrated Supply Chain																	✓			✓								
Adjust (reconfigure) SC design*	✓			✓	✓	✓		✓										✓	✓		✓				✓		✓	
Standardization		✓																✓				✓						
Increase agility	✓				✓	✓		✓	✓					✓				✓			✓				✓			
Sustainability	✓																				✓				✓			
Trust among members	✓							✓	✓					✓							✓							
Risk management culture	✓					✓					✓										✓				✓			
Predictive capability of risk	✓			✓					✓					✓											✓			
SC coordination /Joint planning												✓										✓					✓	
Flexible process (Flexibility)	✓	✓	✓								✓	✓			✓	✓			✓		✓		✓	✓	✓	✓	✓	✓
Corporate social responsibility									✓																			
Information sharing*	✓								✓			✓	✓		✓				✓	✓	✓				✓	✓	✓	
Postponement		✓					✓			✓	✓	✓			✓	✓	✓					✓	✓	✓				✓
Economic supply incentives																							✓		✓		✓	
Concurrent processes													✓				✓										✓	
Continual risk analysis									✓																		✓	
Scenario planning								✓			✓		✓															
Reachability (Cascading)analysis				✓																								
Reducing bullwhip																	✓											
Cross-Trained employees (HR)												✓					✓					✓					✓	
Innovation	✓																								✓			
Risk sharing contracts	✓								✓		✓																✓		✓
Public-private partnership	✓																											
Insurance											✓						✓					✓						
Asset sharing							✓																		✓			
Contingency Planning*	✓							✓		✓	✓				✓			✓			✓			✓				
Re-routing*	✓																							✓				
Speculation*																		✓										
Product bundling*																		✓								✓		
Dynamic pricing*											✓							✓										
Demand switching*											✓							✓						✓	✓			
Operation rerouting*											✓							✓							✓			
Disaster management*											✓							✓										
Supplier evaluation/monitor															✓											✓		
Make and buy		✓													✓				✓			✓				✓		
Flexible supply contracts													✓			✓	✓						✓	✓			✓	
Adding capacity			✓	✓			✓								✓		✓											
Inventory management (VMI)			✓					✓					✓		✓						✓			✓				
Strategic stocking / buffering		✓						✓		✓	✓			✓	✓	✓	✓				✓	✓			✓	✓		✓
Safe location selection																					✓							
Supply base flexibility		✓													✓	✓	✓				✓	✓						
Multiple sourcing	✓	✓	✓			✓	✓	✓		✓	✓	✓	✓	✓					✓		✓		✓	✓			✓	✓
Portfolio diversification								✓			✓				✓									✓	✓			
Collaboration	✓				✓				✓			✓	✓	✓		✓	✓		✓	✓				✓		✓	✓	✓

Note(s): RM enablers marked with * are reactive type

Table 2. Literature summary of various types of RM enablers/practices

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Table 3. Classification and description of RM strategies

RM Strategy	Description	References
Ignore	• Very unusual and rare in reality • When all the possible risk (triggering events) cannot be identified or studied and the impact of risk is too infinitesimal • It should not be confused with risk *acceptance*	Aqlan and Lam (2015) Ceryno *et al.* (2015) Fazli *et al.* (2015) Kersten *et al.* (2007)
Control	• Suitable for operational or day-to-day minor issues (risks) • The impact of risk reduces, but not eliminated completely • Examples include, more quality inspection, better operator training, increasing stock pile, etc.	Jüttner *et al.* (2003) Lavastre *et al.* (2012) Manuj and Mentzer (2008) Giannakis and Papadopoulos (2016)
Accept	• When cost of acting against (managing risk) is more than its impact • Resilience action plans shall be developed well in advance • It's sometimes called risk exploitation or selective risk taking (bearing)	Norrman and Jansson (2004)
Mitigate	• Sophisticated plans designed to counteract against the impact of risk • It should not be confused with risk control • Examples include, make or buy analysis, contract manufacturing, cross training of employees, etc.	
Transfer	• Impact of risk is shared by all entity of the SC • Suitable for disruption risk (very high impact) like natural disaster, terrorist attack • Example include, insurance, revenue/ risk sharing contracts, etc.	
Avoid	• To identify the root cause of (triggering events) of risk and eliminate completely • Residual risk is minimum • Examples include, redesign of SC, (modern) technology adaption, etc.	

alternative outputs which satisfies a set of uncertain input environment (Ganesh, 2006). Oner and Oztaysi (2017) used a FIS model to minimize the risk factors and hence maximize the overall performance of an e-commerce SC. Singh *et al.* (2018) employed a hybrid method of FAHP and FIS to evaluate the sustainability of manufacturing SMEs. Carrera and Mayorga (2008) used a modular FIS for selecting a supplier for new product development. Pourjavad and Shahin (2017) used FIS to evaluate the green SC performance for paint coating industries in Iran.

Table 4 demonstrates the comparison of the current study with some recent ones.

3. Proposed DSS for SCRM

The ultimate objective of this study is to build a DSS containing all the methods discussed in the previous section to assist the SC managers in the assessment of RMSs and the implementation of necessary RMEs. The proposed DSS framework is explained in Figure 1.

We started exploring the RMEs from the systematic literature review by the three-level search criteria mentioned in section 2.2.1 and listed the RMEs. Once the RMEs are listed down, they are validated to ensure the significance of each RME on reducing the overall impact of SC risk. An expert panel, consisting of SC practitioners from heterogeneous industries and academicians are formed. The interview invitation was initially sent to 30 experts located across the country, based on the authors' personal and professional contact by the convenience sampling procedure. After three weeks, email and telephonic reminder requests were sent to the participants. Finally, 12 respondents agreed for the

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	Prior research										
Attributes	Dey (2001)	Faisal *et al.* (2007)	Kumar and Viswanadham (2007)	Micheli *et al.* (2014)	Aqlan and Lam (2015)	Rajesh and Ravi (2015)	Wang *et al.* (2017)	Qazi *et al.* (2017)	Qazi *et al.* (2018)	This research
RM enablers		✓	✓	✓		✓	✓		✓	✓
Evaluation of RM strategies	✓				✓		✓	✓	✓	✓
Relation between RMSs and RMEs										✓
DSS	✓		✓	✓			✓			✓
Techniques	AHP, Decision tree	ISM	CBR	Stochastic LP, Fuzzy Theory	Bow-Tie analysis, LP	Grey DEMATEL	Fuzzy MCDA	FMEA, BBN, Game Theory	FTA, BBN, EUT	CA, FIS, DSS

Table 4. Comparison of current study with the prior studies

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interview. The response rate of 40% provides a good indication for a positive assessment of the survey (Malhotra and Grover, 1998). Twelve experts are deemed an acceptable number for a first-round but further surveys will be conducted in the future to ensure the validity. The demographic background of each respondent, their professional experience, industry categories and ownership structure of the industry are summarized in the appendices (Table A1).

In the primary phase, the questionnaire was emailed to participants, where it was asked to identify only significant RMEs and RMSs, given the choice of "Yes" or "No" for each item. The primary survey aims to ensure the significance of each RMEs and RMSs and reduce repetition and duplicate entries. For instance, RMEs such as *aggregate demand* and *pool demand*; and RMSs such as *risk-sharing*, *risk transfer* and *risk hedging* are merged and considered once. It resulted in a total of 62 RMEs (listed in Table 2) 6 RMSs (listed in Table 3).

The next stage pertains to the mapping of finalized RMEs to the corresponding RMSs. For this, we have restricted the RMEs which are significant and relevant to the case company (explained in section 4.1) and carried out the analysis of CA. The results obtained from CA are used to categorize the RMEs according to various RMSs.

As shown in Figure 1, the final task is to assess (predict) an effective RMS and hence implement the corresponding RMEs. The selection of an RMS primarily depends upon the criticality of the *risk score* (both probability and severity of the risk event). We extend this by

Table 5. Variables and their membership functions

Variable type	Variable name	Linguistic value	Numerical range
Input variables	Risk score	Acceptable	[0 0 0.5]
		Unacceptable	[0.3 1 1]
	Time period of implementation (in months)	Short-term	[0 4 8.5]
		Long-term	[6 11 15]
	Resource availability	Low	[0 0 0.4]
		Medium	[0.15 0.5 0.85]
		High	[0.6 1 1]
Output variable	RM strategy	Ignore	[0 0 0.15]
		Control	[0.10 0.24 0.38]
		Accept	[0.27 0.41 0.55]
		Mitigate	[0.45 0.58 0.72]
		Transfer	[0.62 0.76 0.9]
		Avoid	[0.8 1 1]

Table 6. Summary of the interview

Risk score	Resource availability	Time period of implementation	RM strategy
Acceptable	Low	Short-term	Ignore
Acceptable	Low	Long-term	Control
Acceptable	Medium	Short-term	Control
Acceptable	Medium	Long-term	Accept
Acceptable	High	Short-term	Mitigate
Acceptable	High	Long-term	Transfer
Unacceptable	Low	Short-term	Accept
Unacceptable	Low	Long-term	Mitigate
Unacceptable	Medium	Short-term	Mitigate
Unacceptable	Medium	Long-term	Transfer
Unacceptable	High	Short-term	Transfer
Unacceptable	High	Long-term	Avoid

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adding two variables, viz. "*resource availability*" and "*time period of implementation.*" The resource availability and time period of implementation of RMEs of a firm dictate the choice of RMS (whether to ignore or mitigate or share it) of a firm (Blackhurst *et al.*, 2005). Since each of the aforementioned variables is subjected to a high degree of subjectivity and hence cannot be expressed precisely. We express each variable in form of linguistic categories, for instance "*Risk score*" has been categorized into two levels, *acceptable* and *unacceptable* according to (their tolerance level) and their membership functions are defined accordingly. The linguistic values and membership functions of other variables are summarized in Table 5. A snapshot of modeling of the various decision variables in the fuzzy logic toolbox (MATLAB) is provided in the appendices section (Table A3).

The next task is to map the linguistic values of the input variables to the output variables, for which fuzzy inference system (FIS) is used. One of the crucial steps of FIS is the deduction and formalization of if-then rules (Ghadimi *et al.*, 2017). We conducted the second phase of data gathering (interview) with the expert panel mentioned in Table A2. This time the respondents were requested for face to face interview/discussion. Each interview lasted for approximately 1 h with each respondent. During the interview, the expert was asked about his choice of a particular linguistic value of RMS (say accept or mitigate) under each varying combination of the input variables. The results obtained from the interview are summarized in Table 6.

From Table 6, 12 *if-then* rules have been formulated with the levels mentioned above of each variable. A sample rule is shown below for the explanation.

If the *risk score* is "Acceptable" and *resource availability* is "*Low*" and *time period of implementation* (of the RMEs) is "*Long-term*" then prescribed RMS should be "*Control*".

We have used Mamdani's compositional rule of inference for building the FIS in the current research. The four major steps of FIS are *fuzzification, knowledge base (rules), fuzzy inference mechanism and defuzzification* (Oner and Oztaysi, 2017). The selection of the target range for each input parameter was a part of the expert interview process. *Triangular* membership functions with linguistic variable "low", "medium" and "high" have been chosen for each input variable. The rules are combined by "*and*", centroid defuzzification method has been utilized to simulate the FIS in *the Fuzzy Logic Toolbox* of MATLAB 2016A (*Academic License*) in a 2.40 GHz processor speed, 8 GB RAM computer.

Considering the above settings of the FIS, we created an interface of a DSS based on the MATLAB platform (of the same configuration), which considers the three input values and prescribes the suitable RMS after processing the input information. The RMEs against each RMSs are stored in local database. The RMEs list is displayed upon suitable selection of the corresponding RMS. Further, the DSS provides the order of implementation of the RMEs based on their distance from the cluster center, i.e. RMS. In other words, an RME which is close to the RMS should be implemented first and so on. The snapshots and the detailed workflow of the DSS are shown in Figures 2.

4. Applications of the proposed methodology

We demonstrated the efficacy of the proposed methodology considering the case of *Indian petroleum SC* (IPSC). In the following section the description and importance of the case is discussed.

4.1 Description and introduction about the case

With a daily demand of 4.1 million barrel, India is the third largest consumer of crude oil and petroleum products globally accounting for 4.5% of world's total oil consumption in 2015. The oil and gas sector in India is among the six core industries of the country, and it contributes about 15% to the overall GDP of the country. (Tarei *et al.*, 2020). The country has

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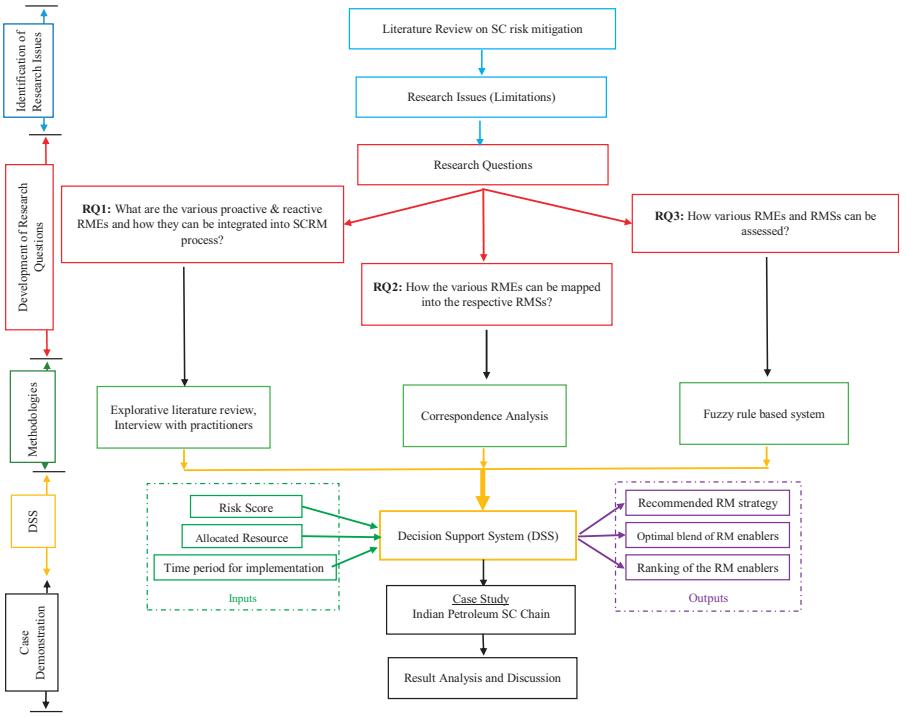

Figure 1.
Proposed research framework

635 million metric tons (MMT) of proven oil reserves (at the beginning of 2015); with almost half of the nation's sedimentary territory is yet to be investigated, the oil and gas part gives enormous opportunities for investment (US EIA, 2017).

Some of the significant risk drivers of (IPSC) are *transportation (delivery system)*, *quality of the petroleum products*, *crude supply*, *customer's order*, *legal/political regulation, economic crisis* and *merger/ alliance* (Tarei *et al.*, 2018). The study mentioned above further reveals that IPSC possesses an overall risk score of *34%*. We extend the existing research by considering the risk score as 34% (as the base one) and simulated two scenarios by changing the rest of the variables (i.e. time period of implementation and resource availability). We evaluate the RMS and the RMEs under these two scenarios in the context of IPSC and present the results.

As mentioned in the previous section (Section 3), we have focused on the case industry and considered the RMEs which are relevant to IPSC for the analysis of CA. We adopted a separate structured interview process focusing on the petroleum SC in the Indian context. A panel of six experts was formed based on their professional expertise in the leading oil and gas industries of the country. The profile of the experts is provided in appendices section (Table A2). The average experience of each member is more than 15 years in the area of sourcing (procurement), planning, refining and marketing (sales). After consulting with the team, total 62 RMEs, which are explored from the literature review, are reduced to 28 due to the uniqueness of Petroleum SC. (Some RMEs like dynamic assortment planning, multiple customer accounts etc. are not relevant to the case industry and hence excluded). In the subsequent step, we ask each expert to link the finalized RMEs to the corresponding RMS based on their expertise (domain knowledge). For instance, if an expert feels that RME: *revenue management* should be implemented, when the overall RMS is planned as *accept*,

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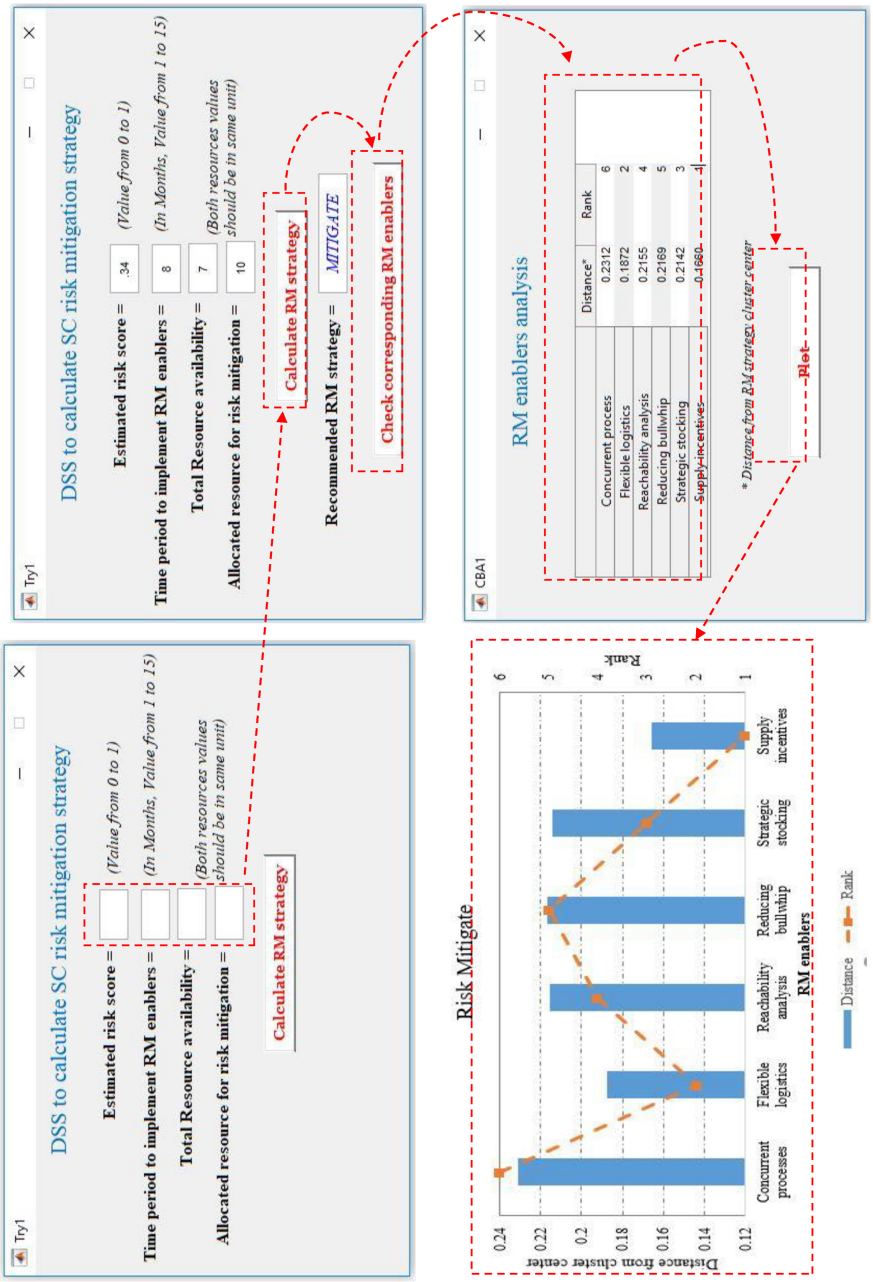

Figure 2.
Workflow of proposed DSS

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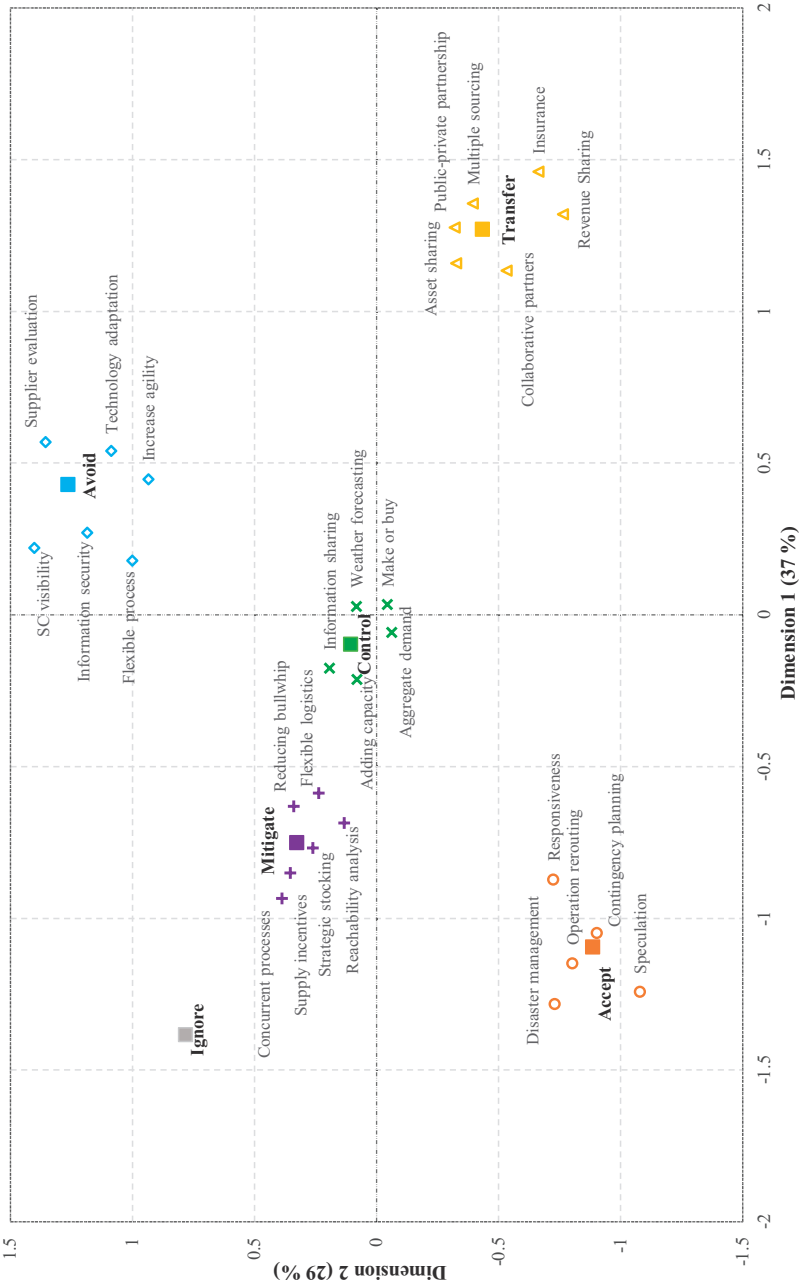

Figure 3.
Relationship of RMEs and RMSs on dimension 1–2 plane

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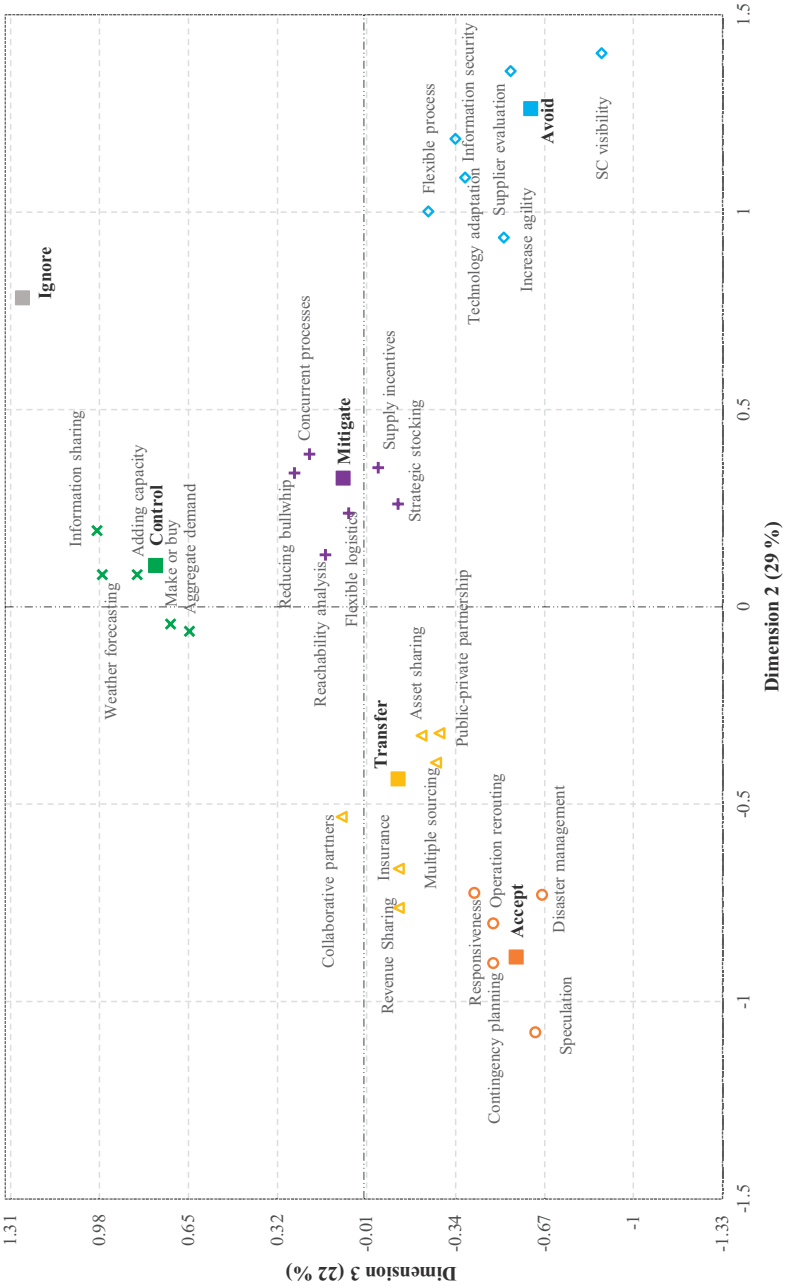

Figure 4.
Relationship of RMEs and RMSs on dimension 2–3 plane

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he/she assigns a score of 1, otherwise 0. The individual response from six experts was collected independently in separate rooms and, they were accumulated to use as input data for analyzing the CA.

4.2 Results of correspondence analysis

The output of the CA is expressed as perceptual maps. The positioning of RMEs on the two-dimensional perceptual map represents the association or belongingness to the closest RMS.

Figures 3 represents the position of various RMEs and RMSs along the dimension 1 and dimension 2 plane. The plane containing dimension 1 and 2 jointly explains 66% variance of the total data. Dimension 1 explains the maximum variance (around 37%) of the data. The figure reveals that *"Accept"* and *"Transfer"* are the two extreme RMS clusters along dimension 1. Risk acceptance is preferred when the SC managers become resilient and ready to face (challenge) the risk, and they intend to bounce back from negative consequence incurred. On the contrary, risk transfer is strategized well beforehand of the occurrence of the risk, which leaves little tasks to be done after the post math. Hence, dimension 1 can be attributed to managers' "*preparedness to face risk*"to deal with risk.

Dimension 2 explains around 29% variability of the total data. As shown Figures 3 and Figures 4 "Accept" and "Avoid" are shown as the two extremes of dimension 2. An organization usually prefers risk acceptance strategy for minor risk events, especially when it realizes that the cost of incurring a loss (due to risk) is less than the cost of preventing the risk. On the contrary, risk avoidance is generally attributed to finding out the root-cause (potential triggering events, which can turn into significant disaster) of risk and eliminating it by changing the (physical) property of the SC. Hence, dimension *2* is regarded as "*Resource Capability*" of a firm to deal with risk.

We have also considered the third dimension 3, which contributes around 22% variance of the total data. As shown in Figures 4 and Figures 5 "Control" and "Avoid" are two extreme RMSs along dimension 3. "Risk *Control*" consists of regular day-to-day operational practices such as, *aggregate* (pool) demand*, weather forecasting*, etc.; whereas "*Risk avoidance*" consists of specific occasional RMEs such as *Speculation*, *Operation rerouting*, etc. Hence, *dimension 3* can be attributed to the degree of "*sophistication*" of RMPs.

Based on the results obtained from CA, and the perceptual maps (explained in the previous sections), a hierarchal taxonomy is proposed (Figure 6). The segregation of RMPs is done, based on the distance of each RMP from the nearest cluster center of RMS positioned on the perceptual map. The taxonomy presents an unblemished representation of the RMSs with their corresponding RMEs.

4.3 DSS application and demonstration:

In order to explain the working principle of the proposed DSS, we have formed and analyzed the following two scenarios in the context of the IPSC.

4.3.1 Scenario 1. In the first scenario, we have considered various input parameters viz. *risk score, time period* and *allocated resources* as *34%*, *8 months* and *7 units* (out of 10), respectively. The DSS recommends "*Risk Mitigate*" as the proper RMS under the given conditions. It further provides the list of RMEs (shown in Figure 7) under the aforementioned RMS. The RMEs are ranked according to their closeness from the cluster (RMS) center.

As shown in Figure 7, the results further indicate that *supply incentives, flexible logistics* and *strategic stocking* are three topmost preferred RMEs under risk "mitigation" strategy. To understand the riskiness of the petroleum SC and to mitigate the negative consequence, the Government of India has taken some policy initiatives recently. For instance, HELP (Hydrocarbon Exploration Licensing Policy, 2016) was launched, which offers various

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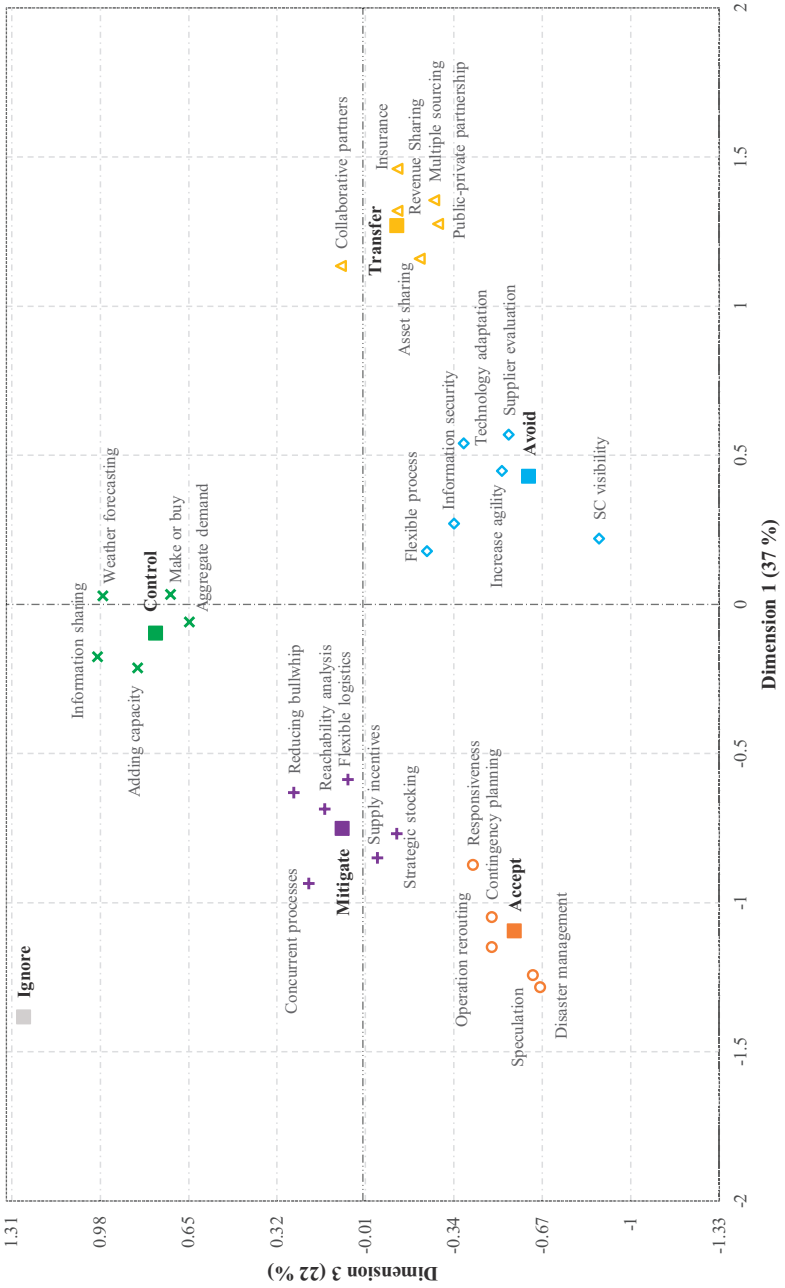

Figure 5.
Relationship of RMEs and RMSs on dimension 1–3 plane

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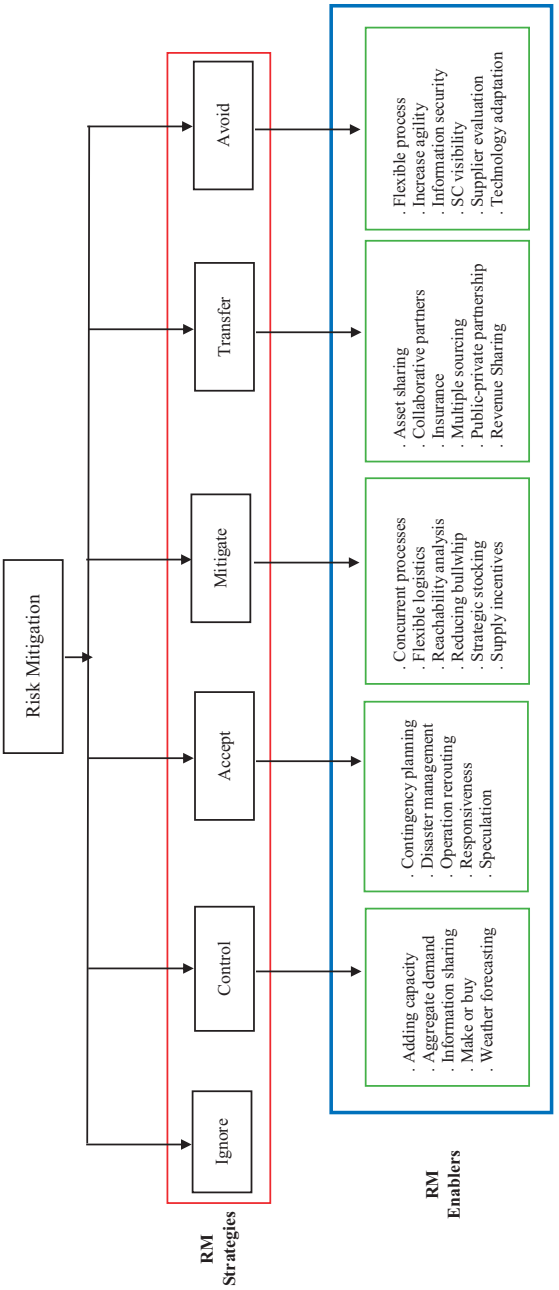

Figure 6.
Hierarchical taxonomy of RMSs and RMEs

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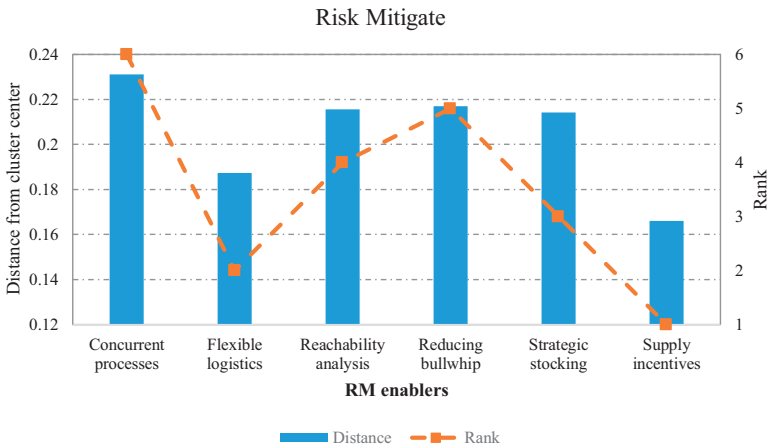

Figure 7. Ranking of the RMEs for risk "Mitigate" strategy

incentives (such as, uniform licensing policy, concessional royalty regime, etc.) to domestic and foreign players in order to make the oil and gas sector of the country conducive to growth. Chabahar port (Iran) was inaugurated on December 2017 with the joint collaboration of Iran, India and Afghanistan. The port opens new (sea) route for transportation of various goods including petroleum (TOI, 2018b). To augment the energy security of India, a joint agreement was signed between Indian strategic petroleum reserve ltd (ISPRL) and Abu Dhabi National Oil company (ADNOC) of UAE to fill 1.5 million tons of *strategic oil reserves* at *Mangalore*, Karnataka (India). (TOI, 2018a). In 2017–18 finance budget, the finance minister of India announced about the setup of two underground petroleum *reserves* at *Chandikhole* (Odisha) and *Bikaner* (Rajasthan), which will increase the reserve capacity of the country to 15.33 tonnes.

4.3.2 Scenario 2. In the second scenario, we have taken the same *risk score* as the previous case (*34%*) but considered a lesser *time period* (*7 months*) and more *allocated resource* (*9 units* out of 10). Under the above-mentioned input parameters, the developed DSS recommends

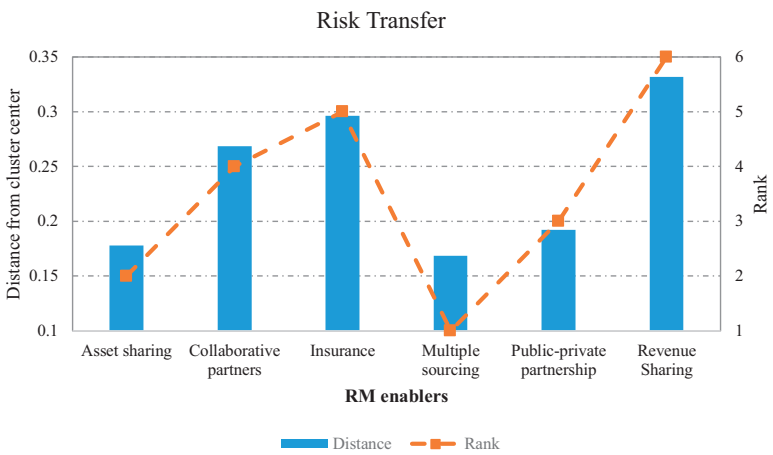

Figure 8. Ranking of the RMEs for risk "Transfer" strategy

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"Risk Transfer" as the suitable RMS. It further lists out seven RMEs (shown in Figure 8) under the risk sharing strategy. Just like the previous scenario, we evaluate the ranking (priority of implementation) of various RMEs.

The ranking of various RMEs are shown in Figure 8. The result reveals that, *multiple sourcing, asset sharing and Public private partnership (PPP) a*s the most preferred overall RMEs. In order to diversify the (crude sourcing) supply base, India had closed a deal with the USA on June 2017. Both state-owned IOCL and private RIL have received an order of crude oil 1.6 million barrels and 1 million barrels respectively on the first week of October last year. (Reuters, 2017). HELP was introduced by GOI to promote the revenue sharing contract. Unlike the earlier *profit-sharing mechanism* (in NELP), the government does not need to scrutinize the *cost* details of the individual players and hence the delays and disputes are avoided effortlessly.

5. Discussion and implication

Irrespective of the case (industry) or the country, SCRM has been recently experiencing global research attention from both practitioner and academician point of interest. The process of SCRM starts from identification and prioritization of critical triggering (disruptive) events and hence, their net (devastating) impact on SC and next, the design of suitable RMSs at the top (strategic) level and implementation of RMEs at the real-world operational level. In this regard, this research work contributes to the existing literature by establishing the mapping (cohesion) between RMSs and RMEs. The relationships (between RMSs and RMEs) are further used to construct a DSS. The DSS considers various critical parameters such *as risk score, the time period to implement the RMEs and resource availability* as the input variables and recommends the *suitable RMS* and corresponding *RMEs* and the *order* of their implementation.

The contributions of this research work are summarized in the following section.

5.1 Theoretical implications

We have adapted various tools from heterogeneous disciplines, viz. CA, a multi-variate statistical extensively used in the marketing and social science research and rule-based FIS to capture the imprecise, vague and uncertain human judgmental information. The proposed methodology is a hybridization of methodologies that does not seem to have been proposed in the domain of SCRM.

We followed a systematic three level keyword search procedure to explore the various literature on SC RMSs and hence the RMEs, irrespective of their nature of implementation (proactive or reactive (resilient)), problem case industry, etc. Additionally, we have presented and justified the definitional (theoretical) distinction between RMSs and RMEs, whereas the existing (prior) literature has experienced the interchangeable usage of the aforementioned terminologies.

The results of CA help us to obtain the underlying relationship between various RMSs. Three underlying factors (dimension) viz. *adaptive capability (courage)* to deal with risk, *short/long term treatment* of risk and *sophistication* of RMEs explain a *significant* proportion of variation of the relationship between RMSs and RMEs.

We have conceptualized twelve rules considering various levels of three essential parameters viz. *risk score*, the *time period to implement* RMEs and *resource availability* for RM and to delineate appropriate *RMSs*. The rules are thereafter fed into a rule-based FIS. The FIS acts as a *knowledge base* to make a decision, i.e. to choose an effective RMS. The FIS additionally takes care of the impreciseness, vagueness and uncertainty of various input parameters in terms of membership functions.

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5.2 Managerial propositions

Based on the results obtained from the proposed research work, we have introduced the following *managerial propositions* (MPs), which will provide critical insights to the domain of SCRM and will assist SC *practitioners* to take the various short term and long-term managerial decisions to deal with SC risk.

MP1. Both RMEs and RMSs hold separate (individual) definitional significance, and they should be designed (or decided) at separate level (hierarchy) of an organization.

An RMS is decided by the top-level executives/managers (at the strategic level), and it shall be aligned with the organization's long-term (goals) mission and vision, competitor's mindset (action), etc. On the contrary, RMEs are the actual practices which are implemented at the bottom (operational) level in order to nullify the negative consequence of risk events (drivers). Once an RMS is decided (selected), the corresponding RMEs shall be designed and assessed (evaluated) according to their preference (order/priority) of implementation. *"Risk management" in an organization is everyone's responsibility. Successful implementation of RM process is a collaborative approach (effort) ranging from people at the strategic level to the operational level*".

MP2. Contextual variables of a firm/organization, like resource availability and time-period of implementation (or planning horizon for implementing RMEs) play a crucial role in SCRM process and hence selecting the proper RMS.

Most of the prior studies focus on the nature of the risk *variables*, like the *likelihood*, or *severity* of risk and hence the *overall risk score* of the SC, while mentioning the RM cycle. The present work, however, considers some additional variables like *resource availability, time-period of implementation (or planning horizon for implementing RMEs)* which influence the decision-making process while selecting the proper RMS and designing corresponding RMEs. "*Resource"* of an industry is a broad term, which ranges from the (technical) knowledge of its employees to financial capability (to hedge against the bad days). "*Time period of implementation"* is the (tentative) planning horizon (short-term/long- term) which a firm takes in order to develop the alternative RMEs, which is further dependent on several factors like organization's core competency, agility, etc. These variables are highly dependent on the context or nature of the industry and they frequently change over time (*dynamic*), The results of this research work shows that inclusion of the contextual variables improves the SCRM process and hence the managers should not ignore such variables and must pay equal attention beforehand,. In addition, the proposed DSS will help to avail (opt) different RMS at different combinations of the aforementioned dynamic input variables.

MP3. The (proposed) DSS is expected to provide the organizations to gain a competitive advantage by expediting the decision-making process in mitigating SC risk, with comparatively more accuracy and at a lesser cost than the conventional methods.

Conventional decision-making is a process of selecting a rational choice from the available alternatives, which is often prone to human (expert) bias, subjectivity, imprecise input data/information, etc. Decision-making in the domain of SCRM strategy selection is not an exception. In this regard, the proposed DSS encapsulates the tacit (unexpressed) expertise knowledge in the form of rules and takes care of the imprecise (incomplete) information by the fuzzy membership functions of the decision parameters. The DSS (after defuzzification) produces crisp outputs, i.e. an RMS and ranks the corresponding RMEs. The proposed FIS based DSS operates with the single click of a mouse and yields results with comparatively less subjectivity bias (more accuracy). However, a wide range of experimental surveys need to be conducted in the future to increase the generalizability and evaluate the benefits of application of the proposed DSS.

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6. Conclusion

SCRM (as a domain) has been experiencing incremental research attention in the recent decades. While, most of the studies have focused on identification and prioritization of triggering events of risk, literature regarding proposing various risk mitigation strategies and practices (enablers) are relatively scanty. In order to address this issue, the present research work has focused on developing a holistic model to predict the suitable RMS under changing risk context and evaluate RMEs. In order to explain the efficacy of the proposed research, we have additionally considered a case study of petroleum SC in the context of a developing country (India). Established tools and techniques from heterogeneous discipline such as marketing research (social sciences) and soft computing are concatenated together to build a DSS. The DSS will facilitate the SC stakeholders and practitioners to make faster decision-making (i.e. assessment of available RMSs) with more accuracy in an uncertain and risky environment.

The prior literature has witnessed the interchangeable usage of RMSsand RMEs. We have differentiated the terminologies in the context of SCRM without compromising with their (terms) etymological definition. The results of our study further conclude that the responsibility of "risk management" in an organization ranges from the strategic level to operational level. The choice of an RMS (in strategic level) not only depends upon risk score but also on some contextual variables like planning horizon and resource availability. Thenceforth, the corresponding RMEs are implemented (in operational level).

Like any other research work, this research also is not free from limitations, which steer the direction for future research opportunities. We have used the expertise knowledge of various SC practitioners from the domestic (developing) country (India) to build the conceptual framework. Hence, although the proposed framework is replicable, scope of generalization of the results is limited. Future research can focus on cross country (both developing and developed) analysis and analyses across industries (manufacturing, process and service). The proposed DSS has the limitation of recommending the RM strategizes and practices for familiar risk events only. The assessment and mitigation of sudden and disruption risk, such as COVID-19 outbreak, are complicated and beyond the scope of the present research. This study, however explicitly focuses on one step of SCRM process, i.e. "Risk Mitigation". The SC managers and stakeholders shall not underestimate the *dynamic* nature of risk. "*Risk Monitoring*" which is a subsequent step of "Risk Mitigation", along with consideration of dynamic nature of risk can be extended as future research work. Hence, the future research work is paramount in the context of SCRM process as a whole.

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Strategy: "a detailed plan for achieving success in situations such as war, politics, business, industry or sport or the skill of planning for such situations:" Retrieved from: https://dictionary.cambridge.org/dictionary/english/strategy

Appendix

Id	Designation	Experience (in years)	Industry	Ownership structure
Resp1	General manager of sales	25	Earth moving equipment manufacturer	FDI (Joint venture)
Resp2	Deputy general manager of manufacturing	20	Automobile manufacturer	Private
Resp3	Project manager	17	Construction	Private
Resp4	Board of directors	28	Oil and gas	Public sector undertaking
Resp5	Vice president	30	Mining	Private
Resp6	Manufacturing head	40	Electronics	FDI (Joint venture)
Resp7	Operations head	37	Petrochemical	FDI (Joint venture)
Resp8	Vice president	40	FMCG	Private
Resp9	Executive director	36	Construction	FDI (Joint venture)
Resp10	Plant head	28	Auto component manufacturing	Private
Resp11	Professor	25	Researcher in the domain of supply chain and operations research	Government institute
Resp12	Professor	35	Academician in the domain of supply chain and operations management	Government institute

Table A1. Profile of respondents for the structured interview

Experts	Designation	Experience (in years)	Industry type
D1	Deputy general manager, sourcing and planning	22	Oil refinery
D2	General manager, sales and marketing	17	Oil refinery
D3	Senior executive, research and development	15	Oil exploration
D4	Divisional manager and sales	24	Oil refinery
D5	Deputy general manager logistics and	22	Oil refinery
D6	Assistant general manager, production planning and control	16	Oil exploration

Table A2. Profile of experts for the case (Oil and Gas) industry

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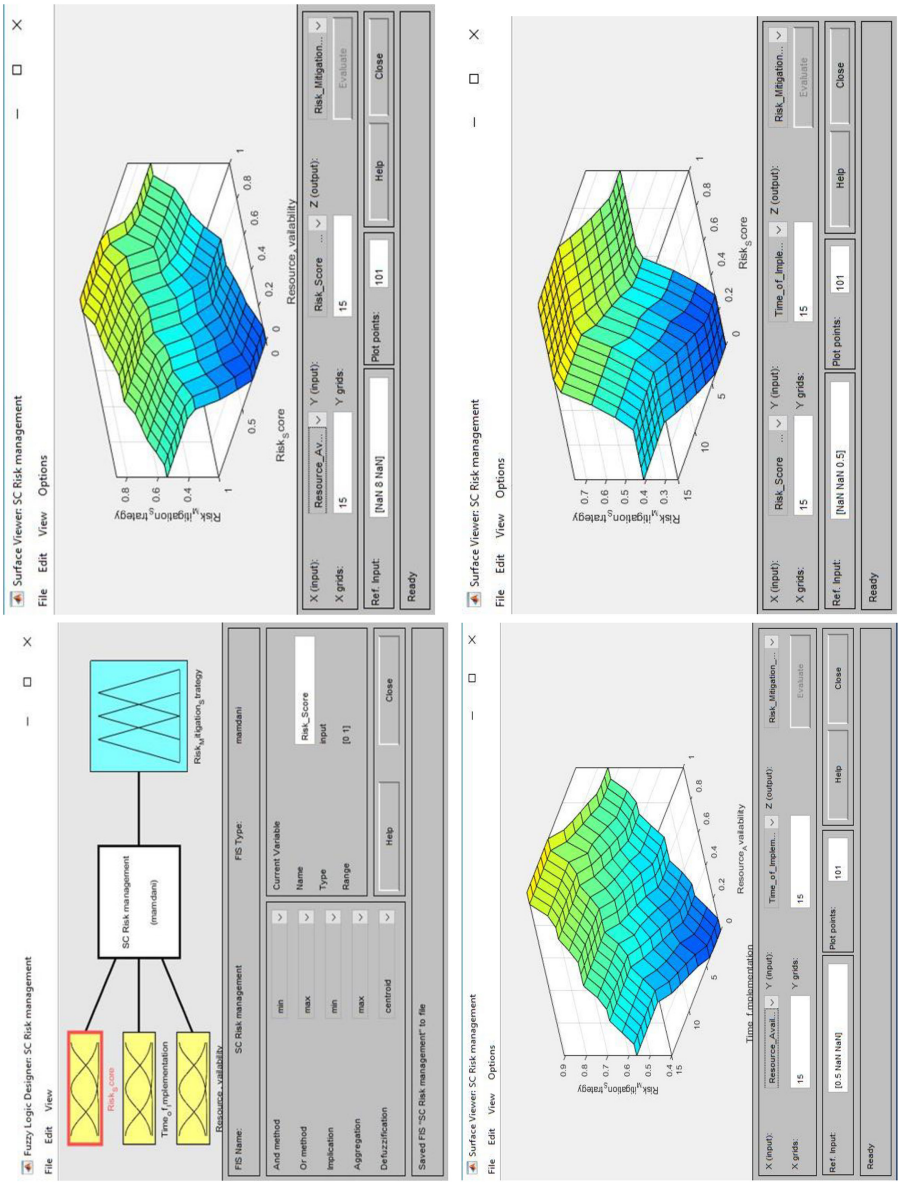

Table A3.
Input-output modeling of the decision parameters in the Fuzzy Logic toolbox in MATLAB

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